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Traditional epidemic models of communicable diseases focus on the temporal dimension. We propose a framework for the epidemic forest approach to spatializing epidemic modeling. An epidemic forest is formed by epidemic trees, each using the tree structure to represent an epidemic starting from a primary case. Each node on the tree represents an individual case, and each link represents a parent–child transmission linkage and can be modeled with the spatiotemporal and other information about the cases. Structural, spatiotemporal, and epidemiological information can be extracted from constructed trees to characterize the epidemic they represent and to correspond to environmental data. When multiple primary cases can be determined, the forest is ready to build. We applied this method to the 2013 dengue fever epidemic in Guangzhou City, China. With the constructed forest, we particularly calculated the case reproduction ratio (R_t), an index widely used for characterizing epidemics, at the global, tree-wise, and pixel-wise scales. We further calculated correlation coefficients between these R_ts and climate variables. R_ts at different scales, as well as their associations with climate variables, offer information at different levels that is all important in epidemiological studies and disease control practices. Through this study, we explored and demonstrated how to spatialize epidemic modeling, what information can be extracted from this spatialization, and then how to use the extracted information. We also point out that spatialization of R_t is the essential process of mapping a communicable disease, corresponding to the spatialization of incidence or prevalence in mapping a chronic disease. *Key Words: Basic reproductive ratio, dengue fever, epidemic forest, Guangzhou, spatial epidemic models.*

传统的传染病之传播模型聚焦时间面向。我们提出传染林方法之架构，空间化传播的模式化。传染林是由传染树所构成，各自使用树的结构来再现从最初案例产生的流行病。树上的每个节点呈现出个别案例，且每个连结皆呈现出家长—孩童的传染连结，并且能够以时空和其他与案例相关的信息进行模式化。结构、时空与流行病学信息，能够从建构的树中取出，以描绘其所再现的特徵，并符合环境数据。当多重原初案例能够被确定时，该森林已准备好进行建构。我们将此一方法应用至2013年中国广州市的登革热流行病。我们运用建构的森林，特别计算案例的再生产率（Rt)，该指标被广泛用来描绘全球、全树以及像素尺度的流行病。我们进一步计算这些Rts和环境变因之间的相关系数。不同尺度的Rts及其与气候变异的关联性，在不同层级提供流行病研究与疾病控制实践中皆相当重要的信息。我们通过此一研究，探讨并展现如何空间化流行病模式化、什麼信息能够从此般空间化中获取，以及如何运用汲取的信息。我们同时指出，Rt的空间化，是绘製可传播疾病的核心过程，并与绘製慢性病的事件或普遍性的空间化相符。 *关键词: 基本再生产率，登革热，传染林，广州，空间传染病模型。*

Los modelos epidémicos tradicionales de enfermedades contagiosas se enfocan en la dimensión temporal. Nosotros proponemos un marco para el enfoque del bosque epidémico para espacializar el modelado epidemiológico. Un bosque epidémico se forma por árboles epidémicos, cada uno de los cuales usa la estructura de árbol para representar una epidemia que empieza a partir de un caso primario. Cada nodo del árbol representa un caso individual, y cada vínculo representa una conexión de trasmisión padre–hijo que puede modelarse con la información espaciotemporal y de otro tipo acerca de los casos. La información estructural, espaciotemporal y epidemiológica puede extraerse desde los árboles construidos para caracterizar la epidemia que ellos representan, y para corresponder con los datos ambientales. Cuando los casos primarios múltiples pueden ser determinados el bosque está listo para construir. Este método lo aplicamos a la epidemia de fiebre de dengue del 2013 en la ciudad de Guangzhou, China. Con el bosque construido,

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particularmente calculamos la ratio caso-reproducción (Rt), un índice ampliamente usado para caracterizar las epidemias a escalas global, en función de árbol y en función de pixel. Calculamos además los coeficientes de correlación entre estas Rts y las variables climáticas. Las Rts a diferentes escalas, lo mismo que sus asociaciones con las variables climáticas, ofrecen información a diferentes niveles, muy importantes en estudios epidemiológicos y en las prácticas de control de enfermedades. A través de este estudio exploramos y demostramos el modo de espacializar el modelado epidémico, qué información puede extraerse de esta espacialización, y después cómo usar la información extraída. Indicamos también que la espacialización de Rt es el proceso esencial para mapear una enfermedad contagiosa, correspondiendo a la espacialización de la incidencia o la prevalencia en el mapeo de una enfermedad crónica. *Palabras clave: bosque epidémico, fiebre dengue, Guangzhou, modelos epidémicos espaciales, ratio reproductiva básica.*

Traditional epidemic models focus on the temporal dimension and are nonspatial; that is, the characterization and prediction are typically about the overall situation of the entire study area, without information about spatial variation. Recently, spatializations of epidemic models have been increasingly attempted and appreciated (Keeling et al. 2001; Viboud et al. 2006; Riley 2007; Balcan et al. 2010; L. Wang and Li 2014). How to achieve this spatialization and what to do with the spatialized models, however, remain questions (Riley et al. 2015).

In studies of communicable diseases, R_t, defined as the average number of infections arising from each new case infected at time t (Riley et al. 2003), is a widely used measurement for characterizing an epidemic process (Dietz 1993; Bacaër 2007; Gubbins et al. 2008). $R_t > 1$ indicates growing infections that demand additional control measures, whereas $R_t < 1$ indicates declining infections (Haydon et al. 2003). Traditionally, R_t is derived from explicit and deterministic susceptible–infectious–recovered (SIR) models and their variants by fitting a series of differential equations to available data (Haydon et al. 2003; Keeling and Rohani 2008). R_t estimated in this way is a simple mean value for the entire study area, without considering the spatial variation that would be important for analyzing the transmission process (Woolhouse et al. 1997). Riley et al. (2015) listed making R_t suitable for spatial models as one of five great challenges for spatial epidemic models.

Haydon et al. (2003) developed a parameter-free method to estimate R_t directly from epidemic data. They constructed so-called epidemic trees for the 2001 UK foot-and-mouth outbreak based on the location and time of each disease case. The nodes on the epidemic trees are the cases of the outbreak. Each link represents a parent–child relationship in the infectious process; that is, the *parent* infected the *child*. Each tree represents the transmission process starting from one primary case. To establish the parent–child linkage, they first identified all candidates to be the parent of a given case: For case a to be a candidate of case b's parent, a must be earlier than b in terms of the onset time and b's onset time must be within the infectious period of a; they then implemented three algorithms to determine the parent, one deterministic and two stochastic. The deterministic algorithm simply chooses, from all candidates, the one that has the shortest spatial distance to the given case to be its parent. The two stochastic ways select the parent based on probability. One considers that all of the candidates have equal probabilities to be the parent, and the other calculates the probability based on the inverse of spatial distance between the two cases. Once the trees are built, R_t can be calculated as the average number of children of all cases whose onset times are within time period t. Haydon et al. (2003) considered that this method is straightforward and robust, taking full advantage of individual-level data; more important, it directly and explicitly characterizes spatial variation in the transmission process and can be directly and explicitly used for analyzing spatiotemporal patterns of that process. Since its publication, however, the method seems not to have been popular for years, likely due to lack of detailed and comprehensive data on disease cases. This situation has changed recently.

Napp et al. (2016) applied the method to calculating R_t of the bluetongue (BTV-1) epidemic in Andalusia, Spain, in 2007. They further spatialized the method in two aspects: First, for the two stochastic methods proposed by Haydon et al. (2003), they set a restriction to only include those cases that are within a certain distance to the child as candidate parents; second, they divided a tree into *foci*, which are subregions of the entire area covered by a tree, and calculated R_t for each focus. The foci were defined based on a subjectively specified threshold in the parent–child distance. It seems that they did this because they did not have data to more accurately determine primary cases that actually started individual epidemics, so they artificially regionalized the model. Once the trees were constructed and R_t values were calculated, they detected associations between R_t and a variety of variables, including those related to the host species and vector species, as well as land cover and elevation at the focus (subregional) level, so that spatial variation of those associations could be revealed.

Lau et al. (2017) developed the method into a more statistically sophisticated and entirely stochastic process that takes into account additional factors, including the incubation period of a case, infectious period of a case, background rate of infection, and demographic features. Although they put their process into the context of identifying transmission linkages on the social network, what they eventually built is epidemic trees. They applied their process to the Ebola outbreak in western Africa (2014–2015) and identified the superspreader phenomenon in the epidemic; that is, a small number of primary cases had played a key role in sustaining onward transmissions and were responsible for a majority of the infections.

Although these previous studies established basic processes for constructing epidemic trees, their calculations of spatialized R_t and other associated spatiotemporal analyses based on the constructed trees are fairly limited and rudimentary. In particular, a connection has not been built between the epidemic tree approach and current literature on disease mapping.

On the other hand, in the literature, most disease mapping studies on communicable diseases simply mimic what has been done with chronic diseases; that is, mapping prevalence or incidence calculated based on population at risk (Johansson, Cummings, and Glass 2009; Althouse, Ng, and Cummings 2011; Colón-González et al. 2013; Struchiner et al. 2015; Wangdi et al. 2018). The purpose of mapping prevalence or incidence is to use these measurements to identify areas that have abnormally high risk (hotspots), and a fundamental assumption underlying the process is that the disease being mapped has a standard rate in the population at risk. This assumption is likely to be invalid for a communicable disease. First, for many communicable diseases, especially acute ones that cause outbreaks, there might not be a rate that can be called *standard* (or *base*), because even one case is extra. Second, the rate of a communicable disease can be hardly a constant in the population at risk but is likely to be dependent on the population density in an area; that is, we expect to see that the rate varies under different population densities. Therefore, the methods and processes for mapping chronic diseases might not be suitable for mapping communicable diseases.

We consider that in mapping a communicable disease, R_t is the counterpart of incidence (or prevalence) in mapping a chronic disease. In particular, the number of parents in R_t is the counterpart of size of population at risk in incidence (or prevalence). Then the task in mapping a communicable disease is to spatialize R_t. In this study, we explored calculation of R_t at three spatial scales, namely, global, tree-wise, and pixel-wise. The pixel-wise calculation of R_t is particularly novel. Essentially, it is a kernel ratio estimation process, a widely used approach in disease mapping (Shi et al. 2013). Through this process, R_t can be spatialized to a high resolution and can be represented by raster data.

We generalize and systematize the epidemic tree method into an *epidemic forest* approach to mapping communicable diseases. We propose a conceptual framework for this approach that comprehensively incorporates the two types of cases that are fundamental in defining the forest—namely, the primary cases and secondary cases—and the three major actors in a transmission process, namely, parent case, child case, and vector (for a vector-borne disease). Each of the three actors has attributes, including incubation period, infectious period, onset time, and location. Attributes of all actors work together to determine the eventual parent–child transmission linkage.

In the next section, we present details of this conceptual framework. We then describe a case study, in which we implemented the epidemic forest approach and applied it to the 2013 dengue epidemic in Guangzhou City, China. We present details of the construction of an epidemic forest using the real dengue case data, characterization of the built forest, calculation of R_t at three different scales, and detection of associations between the epidemic and environmental factors (climate variables) at the three scales. The entire process presented here is implemented by ArcHealth, a software package that can be obtained from the authors.

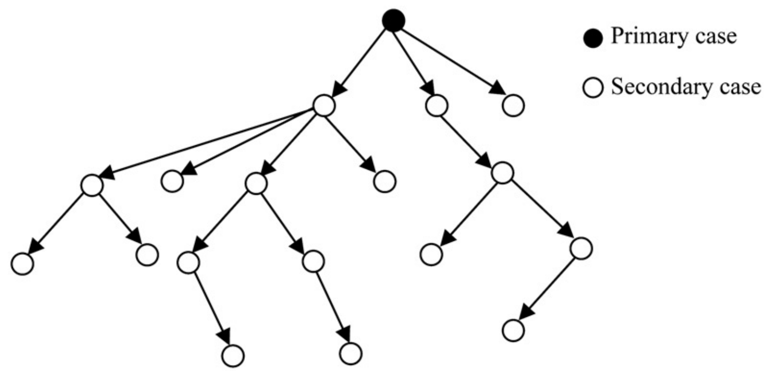

Figure 1. An illustration of an epidemic tree.

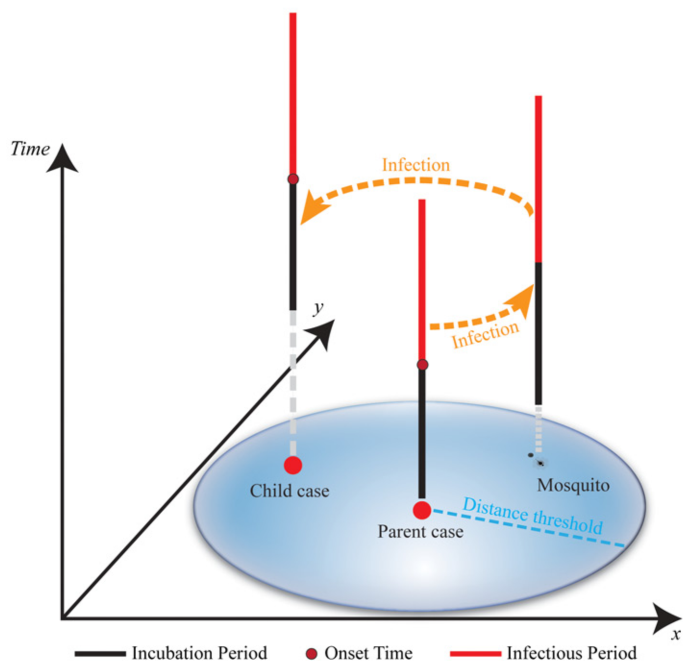

Figure 2. Spatiotemporal relationship of the three major actors in a parent–child infection linkage.

A Conceptual Framework for the Epidemic Forest Approach

Figure 1 illustrates an epidemic tree. A *link* on the tree connects two *nodes*, a *parent* and a *child*. The parent case infects the child case. The root of a tree is called the *primary case*, the case that starts a disease spread process. The other cases on the tree are called *secondary cases*, the cases that are descendants of the primary case in the spread process. We use this structure to model and represent each individual transmission between two cases and the entire spread. When there are multiple primary cases, we will have an epidemic forest. From this representation we can extract information about the epidemic at different spatial scales and then use this information to detect disease–environmental associations and guide interventions.

Here we want to explicitly clarify the distinction between a primary case and a parent case. A primary case is the root of a tree. In other words, it is the ancestor of all of the other cases on the tree (these other cases are called secondary cases). A parent case is a case that has infected others, so all of the cases on a tree, except those "leaf" cases (at the end of branches and without children), are parents of someone else.

Construction of an Epidemic Forest

Construction of an epidemic forest requires individual-level data on disease cases. The process of this construction is based on our understanding of the actual transmission process. The parameter setting in this process requires knowledge of the disease, population at risk, and study area. Essentially, this is a process of identifying the parent for each and every secondary case, based on the spatiotemporal relationship of the three major actors involved in a transmission process, as illustrated in Figure 2.

For a given child case, this parent-identifying process includes three general steps: identifying candidates to be the parent, evaluating the strength of linkage between the given case and each of the candidates, and determining the parent. Once every secondary case finds a parent, the forest is built. We use the concept of forest to generalize the situation. It is certainly possible to have a single-tree forest (i.e., only one primary case eventually became a tree). The three steps are detailed as follows.

Identifying Parent Candidates

In epidemiology, the incubation period is the duration that the subject (patient or vector) has already been infected but has not shown symptoms and usually cannot infect others; the infectious period is the duration during which the subject shows symptoms and can infect others; and the onset time is the starting point of the infectious period. In this study, we measure these two periods by number of days. Thus, theoretically the onset time of a child case (T_c) should fall into a range defined by the incubation and infectious periods of the three actors:

$$T_c \in [t_1,\ t_2], \tag{1}$$

where the range defines the earliest and latest possible time points for T_c:

$$t_1 = T_p + C_v + C_c \tag{2}$$

$$t_2 = T_p + F_p + C_v + F_v + C_c, \tag{3}$$

where T_p is the onset time of the parent case; C_v and F_v are the incubation period and infectious period of the vector, respectively; F_p is the infectious period of the parent; and C_c is the incubation period of the child. For the purpose of identifying parent candidates, we transform these equations to be about T_p:

$$T_p \in [t'_1, t'_2] \tag{4}$$

$$t'_1 = T_c - F_p - C_v - F_v - C_c \tag{5}$$

$$t'_2 = T_c - C_v - C_c. \tag{6}$$

Spatially, we can also set a distance threshold d (Napp et al. 2016):

$$D_{p-c} < d, \tag{7}$$

where $D_{p\text{-}c}$ is the geographic distance between the two cases, which should be determined by the spatial ranges of patients' and vectors' activities during their infectious periods. Equations 4 through 7 can then form a spatiotemporal filter for selecting parent candidates for a given child case.

Under a deterministic method, all of the parameters in the spatiotemporal filter, including all of the incubation and infectious periods, as well as the distance threshold are constant values specified by the researcher. In the real world, it is not likely that all of those parameters hold fixed values; rather, they might follow certain probability distributions. For example, Lau et al. (2017) assumed that in the Ebola epidemic they were studying, the patient's incubation period follows a gamma distribution and the infectious period follows an exponential distribution. Hence, Equations 5 through 7 can take the stochastic form:

$$t'_1 = T_c - f_{Fp}(\cdot) - g_{Cv}(\cdot) - h_{Fv}(\cdot) - i_{Cc}(\cdot) \tag{8}$$

$$t'_2 = T_c - g_{Cv}(\cdot) - i_{Cc}(\cdot) \tag{9}$$

$$D_{p-c} < k(\cdot), \tag{10}$$

where f, g, h, i, and k represent different probability distributions and each could have parameters that need to be estimated.

Evaluating Strength of Linkage between Cases

Once the candidates are identified, we will evaluate the strength of linkage between the child case and all of its candidate parents. Most generically, this strength is evaluated based on the *spatiotemporal distance* between the two cases. The underlying assumption is that the spatiotemporally closer the two cases, the more likely they have a transmission linkage. In this study, we calculate the spatiotemporal distance ($D_{s\text{-}t}$) as

$$D_{\mathrm{s-t}} = \ (1 - W_{\mathrm{t}})\ D_{\mathrm{s}} + \ W_{\mathrm{t}} D_{\mathrm{t}}, \tag{11}$$

where D_s and D_t are the spatial distance and the time difference between the two cases, respectively. W_t is the *weight* assigned to the temporal component in the calculation. Apparently, this linear combination is a simplistic (but flexible) way to integrate the spatial and temporal dimensions. Incorporating the time difference and quantifying its influence in the evaluation of linkage strength between cases are based on the understanding that for a communicable disease like dengue fever, from which most patients would recover, the infectious power of a case is likely to decrease over time.

The spatial distance and time difference are of different units (e.g., km vs. day). To make them compatible for the sake of integration, we use the Z score to normalize them:

$$D_s = D_{coor}/S_{D-coor}, \tag{12}$$

where D_{coor} is the distance calculated with the coordinates, and $S_{D\text{-}coor}$ is the standard deviation calculated based on distances of all possible pairs between cases and their candidate parents; similarly:

$$D_t = D_{date}/S_{D-date}, \tag{13}$$

where D_{date} is the difference between the onset dates of the two cases, and $S_{D\text{-date}}$ is the standard deviation calculated based on the onset date differences of all possible pairs between cases and their candidate parents.

The most important parameter in the spatiotemporal distance calculation is W_t. Under the deterministic method, the researcher specifies a fixed value to this weight. We can also make it stochastic by assuming it follows a probability distribution:

$$W_t = w_t(\cdot). \tag{14}$$

In addition to the spatiotemporal distance, the evaluation of the linkage strength can take into account other factors, such as the social network, transportation, population mobility, and public health interventions. Technically, the information about these other factors can be represented as a matrix, with each element being the value measuring the linkage between two cases. This matrix can then be integrated with the matrix of the spatiotemporal distance to get the overall score of linkage strength for each pair of cases.

Determining the Parent

The eventual parent of a child case will be chosen based on the strength of lineage between the child case and each of its parent candidates. The deterministic method will simply choose the candidate that has the shortest spatiotemporal distance to the child case. A stochastic way will first convert the strength value into probability and then pick the parent based on the probabilities of the candidates. The simplest conversion is based on the inverse distance (Haydon et al. 2003).

Extraction of Information from Epidemic Forest

In this subsection, we outline three types of information that can be extracted from the epidemic forest, namely, structural, spatial and temporal, and epidemiological information. We provide examples of the information in the next section, where we describe the case study.

Structural Information

In terms of graph theory, an epidemic tree is a directed rooted tree (Williamson 1985; Mesbahi and Egerstedt 2010), for which we can use certain indexes to quantitatively characterize the tree's structural properties (Karl and Danscoine 1989) to obtain knowledge about the disease spread. At the node level, the two most important graph indexes are depth and degree. The *depth* of a node is the generation of the node since the root, which tells the position of the case in the transmission process. The *degree* of a node is the number of children the node has, a measurement of infection ability of a case.

At the tree level, important indexes include number of cases, height, and statistics of degree of a tree (mean, maximum, and standard deviation). The number of cases on a tree is a direct measurement of how capable a primary case is in spreading the disease. The height of a tree is the total number of generations on a tree. Statistics of degree for a tree can measure the overall quantity and dispersion of children across nodes. For example, a small standard deviation indicates that the nodes tend to have similar numbers of children.

Basic Spatial and Temporal Information

The epidemic tree can also tell basic temporal, spatial, and spatiotemporal information about an epidemic. In the temporal aspect, we can characterize a tree by its starting date (the onset day of the primary case) and ending date (the onset time of the last secondary case on the tree). These two dates define the period of a spreading process starting from a primary case. In the spatial aspect, the properties of an epidemic tree might include geographic coverage, shape of the coverage, and overall transmission direction.

Geographic coverage of an epidemic tree refers to the territory of the tree, in other words, the geographic area covered by the tree. Technically, it can be calculated as the convex hull that encloses the locations of all cases on the tree or as the dissolved buffers around the cases' locations. The overall transmission direction of a tree can be measured by mean and standard deviation of directions of all parent-to-child links on the tree. Mean direction indicates the general spreading direction of the epidemic modeled by the tree, and standard deviation of direction measures how diverse the directions of parent-to-child links are. Here the mean and standard deviation are specially calculated for a cyclic measurement. The standard deviation varies from 0 to 1, with 1 indicating that all transmissions are concentrated toward a single direction and 0 indicating that the transmissions evenly point to all directions; that is, they are mostly dispersed. All spatial properties described here might vary along with the time, giving spatiotemporal characteristics to the tree.

Epidemiological Information

The epidemic forest provides a straightforward way to calculate an important epidemiological measurement, R_t. Most important, it provides the possibility of calculating R_t at different spatial scales (i.e., spatializing R_t). For a particular period t, R_t is defined as the average number of child cases arising from a parent whose onset time is within t (Riley et al. 2003):

$$R_t = C_{pt}/P_t, \tag{15}$$

where P_t is the number of parents whose onset dates are within t and C_{pt} is the number of all children of those parents. Here in we also use P_t and C_{pt} to refer to those parents and children, respectively. Traditionally, R_t is only calculated for the entire study area (i.e., at the global scale), and C_{pt} is obtained by solving differential equations. With the epidemic forest, once the parent cases of t are identified (according to their onset days), the number of children is immediately known and thus their calculation of global R_t is much more straightforward.

With the epidemic forest, R_t can also be calculated for individual trees and even for specific locations. At the tree scale, P_t and C_{pt} are from the tree under concern, and the calculated R_t can be mapped along with the expansion of the tree. If we calculate R_t for each and every location, we eventually can create a continuous surface of R_t represented by a raster data layer. At this pixel scale, the R_t is about each pixel, P_t are from the neighborhood of the pixel, and C_{pt} are all of the children of P_t. Conceptually, the calculation of pixel-wise R_t is a process of *kernel ratio estimation* (Shi et al. 2013). Technically, this calculation can be implemented with the density tool provided by geographic information systems (e.g., ArcGIS 10.4 2016) in three steps: First, we generate a raster layer of P_t by calculating the kernel density of all cases with onset times that are within period t, during which we treat each case as a simple *event point* (i.e., without a quantitative value); second, we generate a raster layer of C_{pt} by calculating the kernel density of the children, still using the cases with onset times that are within t as input, but this time treat each case as a *value point*, using its number of children as the *value*; and, finally, we divide the density surface of C_{pt} by the density surface of P_t to get a continuous surface of R_t. The bandwidth in these kernel density operations defines the neighborhood of the pixel, which should be specified according to knowledge of how far the infectious influence of a case can reach.

Utilization of the Extracted Information

The information extracted from the constructed forest can be directly used to analyze and (geo)visualize the characteristics of the epidemic. Furthermore, the information can be used to detect disease–environment associations and conduct scenario analysis to support intervention planning. Because R_t can be spatialized with the epidemic forest, the detection of disease–environment associations can then also be spatialized, which will take advantage of increasingly available high-resolution environmental data and identify spatial variation in the associations. For example, the pixel-wise R_t can be directly linked to land cover data derived from satellite images and climate data interpolated from meteorological station records. The spatialized detection of disease–environment associations should be able to greatly enhance our ability to model and even predict epidemics.

With the epidemic forest, we can also model the effect of different intervention measures. For example, we can evaluate the outcome of a measure that is particularly applied to a local area at a certain time point by pruning certain branches of the trees. This type of modeling should be helpful to identify the optimal location and time for an effective intervention measure.

Case Study: 2013 Dengue Fever Epidemic in Guangzhou, China

Dengue fever is a vector-borne tropical infectious disease transmitted by mosquito species *Aedes albopictus* and *Aedes aegypti* (Gubler 2002). In the past fifty years, dengue has become the most prevalent vector-borne disease globally, evolving from a sporadic disease to a major public health problem (Guzman and Harris 2015). According to the World Health Organization (WHO), more than 3.9 billion people in more than 128 countries are at risk of contracting dengue, with 96 million cases estimated per year (WHO 2018b). In China, dengue has been a rapidly growing epidemic and culminated in 2014 with an outbreak of more than 46,000 cases nationwide.

Dengue fever is a climate-sensitive disease (Xiang et al. 2017). Studies have found associations between dengue outbreaks and climate factors such as temperature, relative humidity, and rainfall (Descloux et al. 2012; Colón-González et al. 2013).

In this case study, we constructed epidemic forests using the individual-level dengue data of the 2013 epidemic in Guangzhou, China. We then tried to characterize the epidemic by analyzing the constructed trees and spatialize the disease–climate associations by corresponding the R_t derived from the trees to the interpolated climate data.

Study Area and Data

Located on the Pearl River Delta, Guangzhou is the capital city of Guangdong Province, the most economically developed region of China (Figure 3). The city has a population close to 13 million, the third most populous in China, and serves as the center of economics, transportation, and manufacturing in southern China. Its population includes millions of immigrants and migrants, mainly from inland China but also from Africa, Southeast Asia, and other areas. A great volume of travelers move in and out of Guangzhou on a daily basis. The city has a humid subtropical climate with a mean temperature of 13.6 °C in January and 28.6 °C in July and an average annual rainfall of 1,700 mm. All of these conditions favor dengue vector growth and epidemic transmission, which caused the

Figure 3. Dengue cases in 2013 in Guangzhou City.

city to suffer the most serious dengue outbreaks in China since dengue fever started to appear in China in 1978 (Wu et al. 2010).

We obtained individual-level epidemic data for the 2013 dengue fever outbreak in Guangzhou from the National Notifiable Infectious Disease Reporting Information System (NNIDRIS). The data include information about gender, age, address, and onset time (Xiao et al. 2016). All cases were recognized as either imported cases or indigenous cases. An *imported* dengue case is defined as being from a person who has traveled to a dengue-affected foreign country with a history of being bitten by mosquitoes within fifteen days before the onset of illness. *Indigenous* cases are those that do not have evidence of being imported (Li et al. 2012). The 2013 epidemic in Guangzhou consisted of a total of 1,270 cases, with 21 (1.65 percent) being imported and 1,249 (98.35 percent) indigenous. They were geocoded to points using their address information. Figure 3 shows that most of the cases are clustered in the downtown area. Temporally, these cases are widely spread, with the first case diagnosed on 2 February 2013, the first indigenous case diagnosed on 14 July 2013, and the last case diagnosed on 9 December 2013. Over 60 percent of the cases were diagnosed in October (Figure 4).

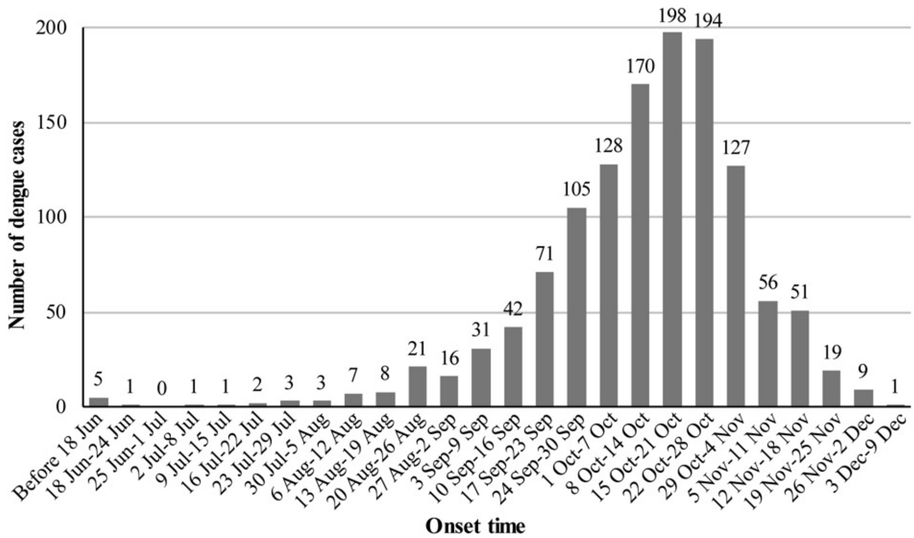

Figure 4. Weekly (seven-day interval) number of dengue fever cases over the course of the 2013 epidemic in Guangzhou, China.

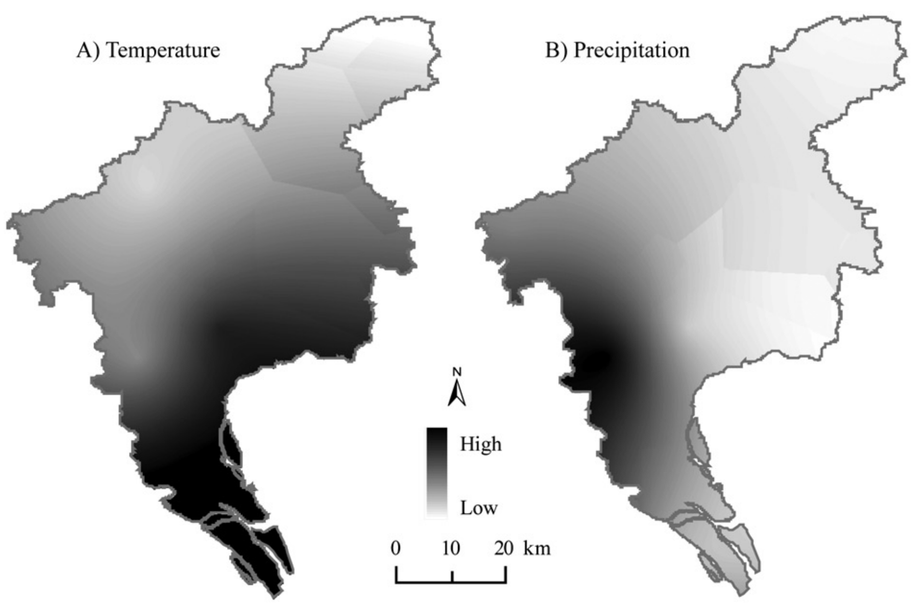

Figure 5. The kriging interpolation results of temperature and precipitation, using the results of the fourteenth epidemic week as examples.

Daily meteorological data for 2013 were collected from the China Meteorological Data Sharing Service System (2017). Data from the twenty-three meteorological stations that are within 150 km of Guangzhou City were selected and put into ordinary kriging interpolations (Holdaway 1996; Vicente-Serrano, Saz-Sánchez, and Cuadrat 2003; Carrera-Hernández and Gaskin 2007) to generate continuous surfaces. The daily data were aggregated to seven-day intervals, corresponding to the weeks defined for the dengue cases based on their onset dates (described later), to detect associations between climate conditions and the dengue epidemic on a weekly basis. Two climate variables tested in this study are weekly average temperature and weekly precipitation. Figure 5 shows examples of the kriging interpolation results of the two climate variables.

Construction of Epidemic Forest for the Dengue Epidemic

Epidemiological analyses confirm that dengue fever is not endemic in mainland China, which means that Guangzhou's dengue outbreaks were due to imported cases (Lai et al. 2015; Sun et al. 2017). This makes determining primary cases straightforward in our study: All imported cases would be roots of potential epidemic trees.

Because this article focuses on formalization of the epidemic forest approach and spatialization of modeling, for this case study we only present the forest constructed with the generic and deterministic algorithm; that is, the trees that were created based on researcher-specified parameter values and used the closest candidate as the parent of a given child case.

Equations 5 through 7 and 11 contain a total of eight parameters for which values need to be assigned. In this study, we did not specify a value for the distance threshold d in Equation 7, because the potential infectious range of a patient in the study could cover the entire city, because Guangzhou is a metropolitan area with well-developed road networks and transportation systems and thus travel between any two given locations in the city is possible. Eventually, we specified values for three parameters: the patient's incubation period and infectious period (applied to both parent and child cases) and the time weight.

Because both parent case and child case are about human patients, it is reasonable to use a single set of incubation and infectious periods for both. According to the literature (Centers for Disease Control and Prevention 2014; WHO 2018a), the incubation period is four to ten days and the infectious period typically lasts three to twelve days. An individual mosquito might have an incubation period of four to twelve days and then remains infectious for the rest of its life, which might be days or a few weeks. The infecting process of mosquitos can be complicated, however, because a transmission might involve more than one mosquito and the virus can be passed on from a mother mosquito to its children (Joshi, Mourya, and Sharma 2002; Arunachalam et al. 2008). Due to this great uncertainty, we consider it not worth the struggle of explicitly specifying values for the mosquito's incubation and infectious periods before more knowledge is acquired. Instead, we extended the infectious period of the human patient to implicitly represent the effect of the vector.

To test the sensitivity of the method to parameter settings, we experimented with different values for each parameter. For the incubation period (IC) of a patient, we tested four, seven, and ten days; for the extended infectious period (exIF) of a patient, we tested twenty, thirty, and forty days; and for the time weight, we tested 0, 0.25, and 0.5. The combination of different parameter values resulted in a total of twenty-seven parameter settings.

Table 1 reports the basic information of the epidemic forests built under the twenty-seven parameter settings. The trees' IDs are ordered according to the onset date of the imported cases. *We have some quick observations. First and most important, in a forest there is almost always a dominant tree that takes more than a half (sometimes above 80 percent) of the indigenous cases, and in sixteen of the twenty-seven forests, this dominant tree is T6. In the other eleven forests, either the dominant tree roots in a different imported case or there are two head-to-head competing trees together taking more than 80 percent of the indigenous cases. T5 is dominant in four of the eight forests of the former, all with $W_t = 0$. T7 takes the other four, all with exIF $= 20$ days. In two of the three two-head situations, T6 is again the leading one. In none of the twenty-seven forests do the indigenous cases evenly disperse across different trees. These observations indicate that the superspreader(s) found by Lau et al. (2017) in Ebola epidemics might also exist in dengue epidemics. The results also show that the two earliest trees in the forest are more likely to become the dominant tree.

Table 1. Basic information of epidemic forests generated with different parameter settings for the 2013 dengue epidemic in Guangzhou, China

					Number of cases in specific tree														
IC	*exIF*	W_t	Starting date	Number of trees	T4	T5	T6	T7	T18	T20	T23	T26	T34	T50	T64	T67	T131	T169	T445
4	20	0	2 Jul	7				696	96		342	33		37	24	28			
4	20	0.25	2 Jul	9				810				25	274	36	12	28	2	69	2
4	20	0.5	2 Jul	7				1,108				26		37	26	28		29	2
4	30	0	12 Jun	9		84	718		90		246	32	25		24	28	11		
4	30	0.25	18 Jun	9			810					25	274	36	12	28	2	69	2
4	30	0.5	18 Jun	7			1,108					26		37	26	28		29	2
4	40	0	2 Jun	11	251	717			89		78	32	23	5	24	28	11	2	
4	40	0.25	18 Jun	9			810					25	274	36	12	28	2	69	2
4	40	0.5	18 Jun	7			1,108					26		37	26	28		29	2
7	20	0	18 Jun	8			684	41	171		257	3		49	24	28			
7	20	0.25	18 Jun	9			949	41	2			3	177	38	18	28	2		
7	20	0.5	18 Jun	8			772	188		2		28		219	118	28	2		
7	30	0	12 Jun	9		2	849		80		222	17	25		24	28	11		
7	30	0.25	18 Jun	8			986		2			6	177	38	18	28	2		
7	30	0.5	18 Jun	7			959			2		28		219	18	28	2		
7	40	0	2 Jun	10	275	729			73	27	75	17		5	24	28	6		
7	40	0.25	18 Jun	8			986		2			6	177	38	18	28	2		
7	40	0.5	18 Jun	7			959			2		28		219	18	28	2		
10	20	0	18 Jun	9			26	699	191	38	238	34		6	10	16			
10	20	0.25	18 Jun	9			35	525	499			30	96	27	12	11		23	
10	20	0.5	18 Jun	10			747	279		19	2	25	130	24	12	11		10	
10	30	0	12 Jun	11		685	184		164	11	116	32	15		24	16	11	2	
10	30	0.25	18 Jun	8			555		499			30	96	27	16	11		23	
10	30	0.5	18 Jun	9			1,025			19	2	25	130	24	12	11		10	
10	40	0	2 Jun	10	287	643			157	27	60	32		7	24	16	6		
10	40	0.25	18 Jun	8			555		499			30	96	27	16	11		23	
10	40	0.5	18 Jun	9			1,025			19	2	25	130	24	12	11		10	

Note: IC = incubation period. T4, T5, …, and T445 are IDs of the imported cases. The ID of cases in the table are ordered according to the cases' onset dates. The results of the shaded settings were used in the following analysis.

The results demonstrate that values of the three parameters do have effects on the resulting forest. Small values of W_t ($W_t = 0$), exIF (exIF = 20 days), or both are likely to lead to forests quite different from those under other settings. Especially when $W_t = 0$ (i.e., the evaluation of the linkage strength considers only the spatial distance), the forest usually has more trees and an earlier starting date than those from its comparable settings. When the parameter values increase, the discrepancies among forests seem to decrease. Especially when exIF $\geq$ 30 days, the results are no longer affected by this parameter. Generally, the results are more sensitive to W_t than to exIF, and then to IC. The sensitivity might level out as the parameter values increase. Figure 6 shows the nine forests generated with IC = 4 days. These maps demonstrate the effects of different values of W_t and exIF.

Not all twenty-one imported cases have trees, and a total of fifteen imported cases have at least one tree under different settings. The number of trees in a forest varies between seven and eleven, and all trees' durations are between early June and early October. The six imported cases that did not get trees under any parameter setting occurred in either early February or November, two of the months that do not favor formation of mosquito habitats, in terms of the climate conditions. The patients of the fifteen imported cases that have trees include nine men and six women, with ages ranging from seventeen to sixty. Most of them lived in the Tianhe, Haizhu, and Liwan districts, the central area of Guangzhou City (see Figure 3). The patient of T6, the superspreader in most forests, is a forty-year-old businessman. T26, T64, and

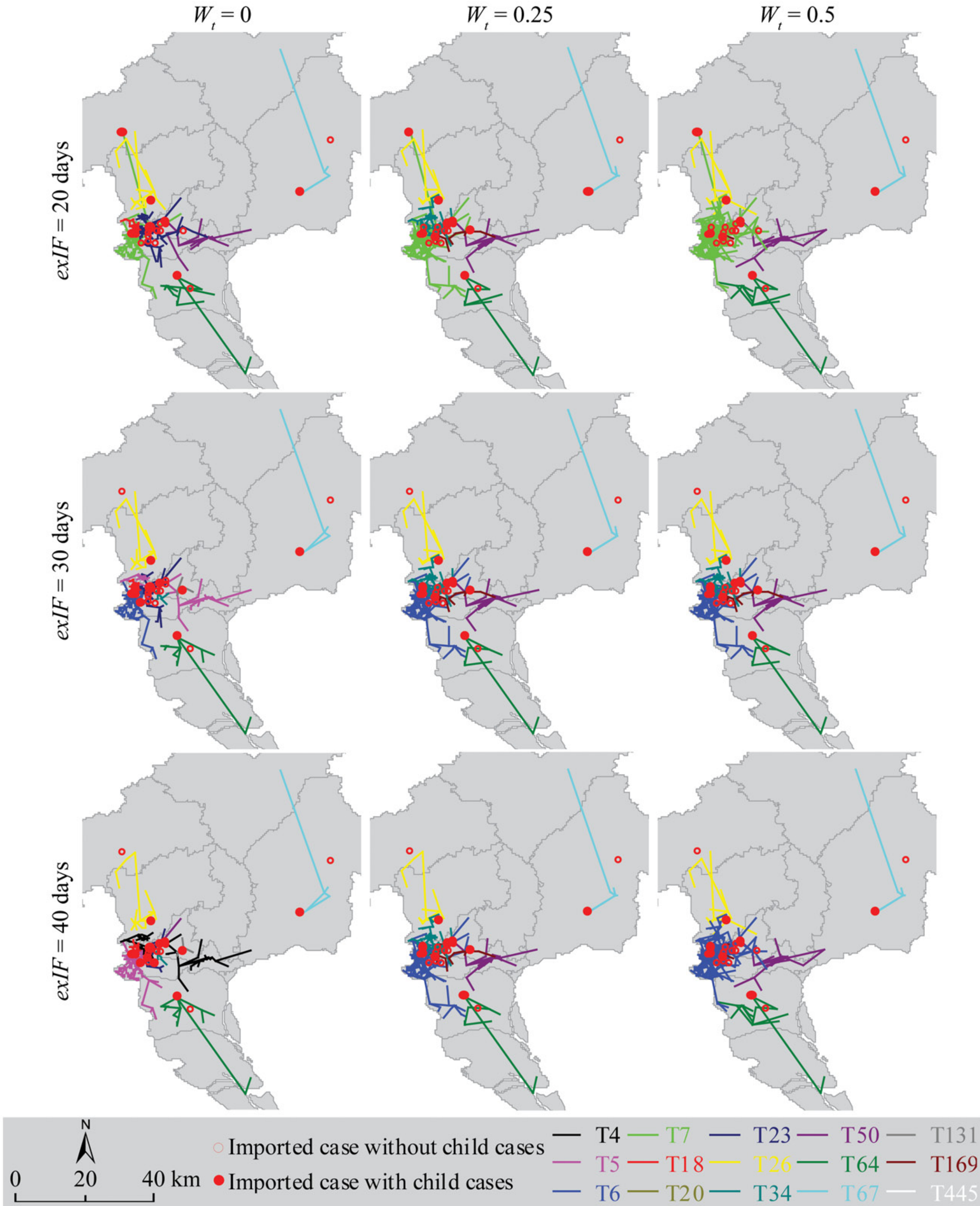

Figure 6. Epidemic forests generated with different values of the time weight (W_t) and extended infectious period. The patient's incubation period is set to be four days. exIF = extended infectious period.

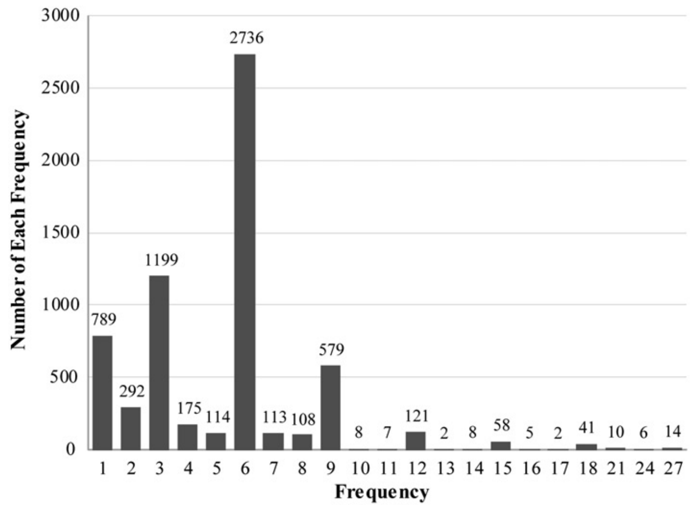

Figure 7. Frequencies of unique parent–child links under twenty-seven different parameter settings for the 2013 dengue epidemic in Guangzhou.

Table 2. Node-level graph features of the epidemic forests of the 2013 dengue epidemic in Guangzhou

Setting 1: IC = 4 days, exIF = 30 days, $W_t = 0.5$					Setting 2: IC = 10 days, exIF = 20 days, $W_t = 0.25$				
Case ID	Parent ID	Tree ID	Depth	Degree	Case ID	Parent ID	Tree ID	Depth	Degree
8	6	T6	1	2	8	6	T6	1	2
9	8	T6	2	2	9	6	T6	1	3
10	8	T6	2	1	10	7	T7	1	3
11	9	T6	3	1	11	8	T6	2	0
12	9	T6	3	2	12	8	T6	2	0
13	11	T6	4	0	13	9	T6	2	3
14	12	T6	4	2	14	9	T6	2	0
15	12	T6	4	0	15	9	T6	2	1
16	10	T6	3	2	16	10	T7	2	1

Note: IC = incubation period; exIF = extended infectious period. Parent ID is the ID of the case that has directly infected the given case. Tree ID is the ID of the tree, which uses the ID of the primary case (the root of the tree).

T67 have trees in all forests, but none of the trees is large, indicating that these three cases are highly certain to have infected other people but their influences might be limited to fairly local spatio-temporal domains.

The twenty-seven forests generated with different parameter settings contain a total of 6,387 unique parent–child links. Figure 7 shows the frequencies of those unique links, which can be used to evaluate the uncertainty of the links. The distribution of the frequencies has a long tail, suggesting a Poisson distribution. More than 40 percent of the links have a frequency of six; only fourteen links have the highest certainty; that is, they occur in every forest.

Properties of the Constructed Epidemic Forests

For illustrative purposes, we selected forests generated with two of the twenty-seven parameter settings for further analyses. The first is IC = 4 days, exIF = 30 days, and $W_t = 0.5$. The forest generated with this setting has the most dominant tree among all settings, which takes 1,108 cases, more than 88 percent of all cases on the trees. The other selected setting is IC = 10 days, exIF = 20 days, and $W_t = 0.25$. This setting's largest tree is the least dominant among all settings, which takes only 525 cases, about 42 percent of all cases on the trees (see Table 1). Here in, we refer to the first setting as Setting 1 and the second setting as Setting 2.

Table 3. Tree-level graph features of the epidemic forests of the 2013 dengue epidemic in Guangzhou

	Tree ID	Number of cases	Height	Degree		
				Mean	Max	*Std*
Setting 1: IC = 4 days, exIF = 30 days, $W_t = 0.5$	T6	1,108	31	0.999	16	1.420
	T26	26	11	0.962	4	1.148
	T50	37	15	0.973	6	1.280
	T64	26	8	0.962	5	1.428
	T67	28	10	0.964	5	1.347
	T169	29	8	0.966	7	1.500
	T445	2	1	0.500	1	0.707
	Overall	1,256	31	0.983	16	1.404
Setting 2: IC = 10 days, exIF = 20 days, $W_t = 0.25$	T6	35	8	0.971	4	1.224
	T7	525	13	0.998	10	1.627
	T18	499	11	0.998	10	1.558
	T26	30	7	0.967	4	1.273
	T34	96	9	0.990	11	1.645
	T50	27	8	0.963	4	1.126
	T64	12	3	0.917	5	1.621
	T67	11	2	0.909	7	2.119
	T169	23	5	0.957	5	1.397
	Overall	1,258	13	0.983	11	1.566

Note: IC = incubation period; exIF = extended infectious period.

As examples, Table 2 shows two graph properties of some nodes from the trees generated with Settings 1 and 2. Most of this same group of cases have different property values under different settings. For Setting 1, all of the example cases are from T6, the dominant tree, and are on the first through fourth generations of T6. For Setting 2, most of the cases are still from T6, but Cases 10 and 16 are from T7. All cases for Setting 2 are either first- or second-generation cases. Among all cases, only Case 8 has the same property values in both settings.

Table 3 lists the tree-level graph properties of the two forests. Under Setting 1, T6 is a superspreader, taking more than 88 percent of the indigenous cases. It is also the tallest tree, with thirty-one generations. Its mean and max degree are also the largest among all trees under this setting. Its mean degree is 0.999 and standard deviation of degree is 1.42, indicating that most cases on this tree have a small number of children (not exceeding two). Under Setting 2, T7 is the biggest and tallest one but is much less dominant, with a second major tree (T18) present.

Table 4 shows basic spatial and temporal properties of the two forests. We delineated the geographic coverage of a tree by creating buffer around each and every case location, with 5 km as the radius. The buffer circles of different case locations were dissolved into a single polygon; that is, the overlapping parts of different buffer circles were merged. We chose to use the buffering method to define the spatial coverage of a tree, because, compared with other methods (e.g., the convex hull), the result of this method can more precisely represent the shape of the tree and the uncertainty of the case location. We subjectively chose 5 km as the buffer radius, considering it as an estimate of uncertainty in the case location data and being suitable for illustration purposes. Under Setting 1, T6, the superspreader, has the earliest starting date, the latest ending date, and the largest geographic coverage among all trees in its forest. Its mean transmission direction is east (80°), with a small standard deviation indicating that the directions on this tree are not concentrated. Under Setting 2, T6, no longer the dominant one, still has the earliest starting date but ended earlier than most other trees. T7 and T18, the two largest and competing trees under this setting, started later than T6 by more than two weeks but lasted much longer. Except for T6, the trees of the other imported cases shared by the two settings, including T26, T50, T64, T67, and T169, maintain considerable consistency between the two forests, especially on the transmission direction. Among them, T50 is least affected by the parameter setting.

Table 4. Spatial and temporal properties of the epidemic forests of the 2013 dengue epidemic in Guangzhou

Settings	Tree ID	Starting date	Ending date	Mean direction	Std of direction	Geographic coverage (km^2)
Setting 1: IC = 4 days, exIF = 30 days, W_t = 0.5	T6	18 Jun	1 Dec	80	0.102	873.637
	T26	15 Aug	8 Nov	179	0.904	551.387
	T50	26 Aug	29 Nov	273	0.884	583.758
	T64	1 Sep	1 Nov	326	0.922	532.887
	T67	2 Sep	2 Nov	250	0.964	279.160
	T169	20 Sep	23 Oct	244	0.527	158.443
	T445	7 Oct	11 Oct	168	0.301	83.947
	Overall	18 Jun	1 Dec	93	0.012	2,070.125
Setting 2: IC = 10 days, exIF = 20 days, W_t = 0.25	T6	18 Jun	2 Nov	243	0.842	612.424
	T7	2 Jul	1 Dec	13	0.187	817.487
	T18	7 Aug	30 Nov	110	0.237	641.344
	T26	15 Aug	25 Nov	184	0.874	581.433
	T34	20 Aug	22 Nov	237	0.549	464.420
	T50	26 Aug	29 Nov	274	0.890	507.757
	T64	1 Sep	28 Oct	321	0.947	438.275
	T67	2 Sep	27 Sep	251	0.947	178.909
	T169	20 Sep	22 Nov	265	0.655	253.429
	Overall	18 Jun	1 Dec	92	0.015	2,085.618

Note: IC = incubation period; exIF = extended infectious period.

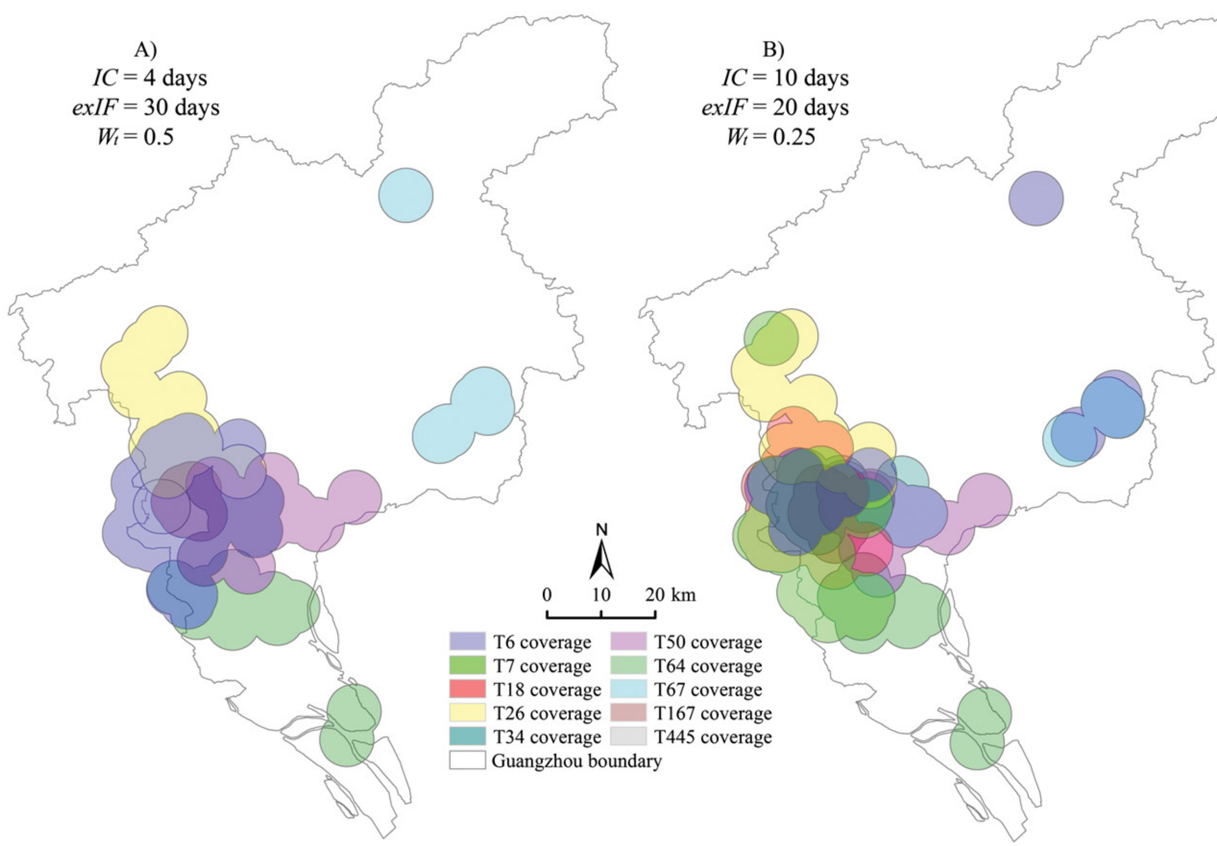

Figure 8. Geographic coverages of different trees in the forests under two different parameter settings. IC = incubation period; exIF = extended infectious period.

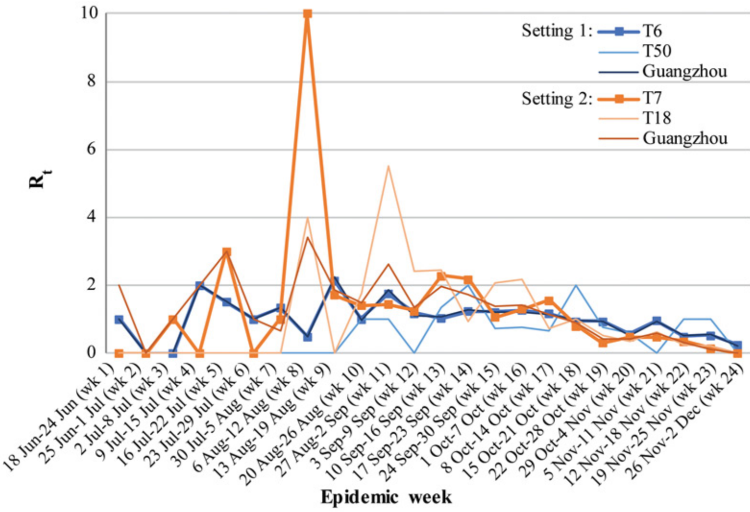

Figure 9. Global R_ts and tree-wise R_ts of the 2013 dengue epidemic in Guangzhou. Here the weeks are not the calendar weeks but seven-day intervals staring from the starting date of the epidemic tree, 18 June 2013. Tree-wise R_ts of four selected trees are shown here as examples.

As shown in Figure 8, the geographic coverages of different trees in a forest have great overlaps (especially under Setting 2), demonstrating that constructing an epidemic forest is an effective way to disentangle different transmission processes that occur over similar but still distinctive geographic areas and facilitate delineation, analysis, and comparison of spatiotemporal domains of individual processes. The size of coverage is far less discrepant than the number of cases across different trees. Some trees with relatively small numbers of nodes might still extend across fairly large geographic areas. For example, under Setting 1, T6 has 1,108 cases and occupies 873.6 km^2, whereas T26, with only twenty-six cases, occupies 551.4 km^2.

With the constructed epidemic forests, we calculated R_t for each epidemic week at three spatial scales. In this study, we define that Week 1 begins on the onset day of the first imported case that has a tree. Both of our sample settings have the same first imported case (T6), the onset time of which is 18 June 2013. Thus starting from this date, we count every seven days as a week, until the week of the last indigenous case (1 December 2013). In this way, we defined twenty-four epidemic weeks for the 2013 dengue epidemic in Guangzhou.

Figure 9 shows the global R_t of Guangzhou City and the tree-wise R_t of a few selected trees as examples. Figure 10 provides a spatiotemporal visualization of the trees with R_ts that are presented in Figure 9. Figure 10 clearly shows that these major trees in their forests have distinctive spatiotemporal patterns. This information, along with the temporal variation of R_t of individual trees shown by Figure 9, allows a spatiotemporal characterization of individual trees. Under Setting 1, the global R_ts are largely determined by the R_ts of the dominant tree. The second largest tree of this setting, T50, which has thirty-seven cases, has an R_t time series quite different from the global one. Under Setting 2, where the biggest tree is less dominant, the tree-wise R_ts of the first and second largest trees are both considerably different from the global ones. Figure 9 demonstrates that tree-wise R_ts can offer information that is not available from the global R_ts.

Figure 11 shows the raster representations of pixel-wise R_t for a few weeks. In this study, for illustrative purposes we subjectively set bandwidth = 5 km to calculate the pixel-wise R_t. Figure 11 demonstrates that R_t values localized down to the pixel level are able to provide highly detailed information about spatiotemporal variation in the disease spread. The maps show that Week 17 (8 October–14 October) is the peak of the epidemic, and Week 22 (12 November–18 November) is close to the end of the epidemic. The maps also show that the general patterns of pixel-level R_t calculated based on the two forests are similar.

Assessment of the Association between Climate Factors and R_t

Previous studies have found that dengue disease is related to climate factors, such as temperature, rainfall, and relative humidity (Depradine and Lovell 2004;

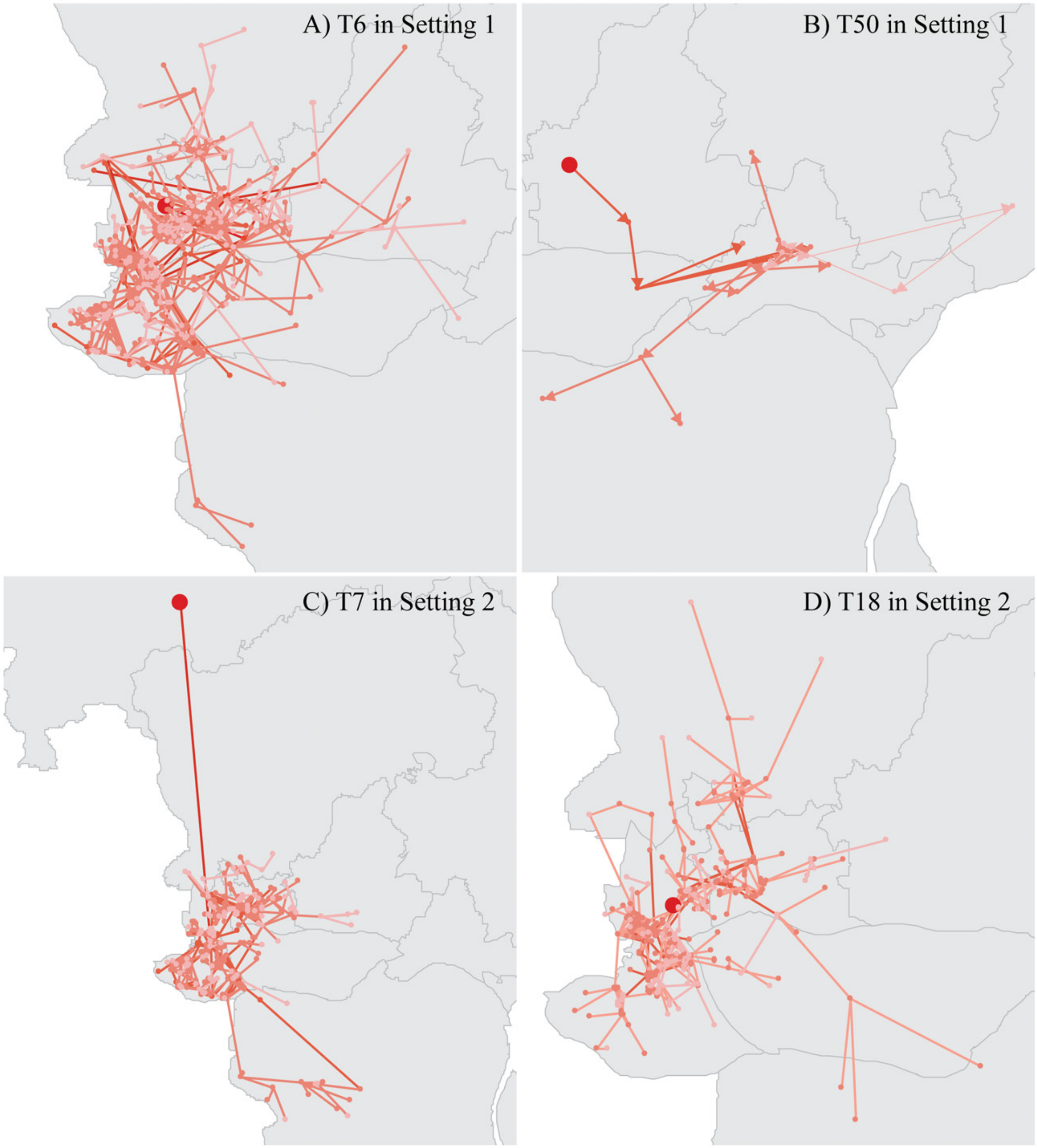

Figure 10. Spatiotemporal distributions of epidemic trees of the 2013 dengue epidemic in Guangzhou. The color of lines denotes the onset time of the child cases. The darker the tone, the earlier the onset date. Four selected trees are shown here as examples.

Bangs et al. 2006; Earnest, Tan and Wilder-Smith 2012; Shen et al. 2015). A lag effect of climate factors on dengue has also been identified (Chen et al. 2010; Chien and Yu 2014; Xiang et al. 2017). Most of those studies assess the dengue–climate association at the scale of the entire study area. Researchers, have pointed out, however, that such associations are site-specific and could vary greatly within the area of a country or even a province (Morin, Comrie, and Ernst 2013; Xiang et al. 2017).

In this study, we chose two climate factors, weekly average temperature and weekly total precipitation, as

Figure 11. Localized (pixel-wise) R_t based on epidemic forests of the 2013 dengue epidemic in Guangzhou. The grayscales of different maps have been normalized and are comparable with each other. The darker color denotes higher value of R_t. Maps of four weeks are shown here as examples. IC = incubation period; exIF = extended infectious period.

Table 5. R_t–climate correlation coefficients at the global and the tree-wise scales for the 2013 dengue epidemic in Guangzhou

		T6		T50		Guangzhou	
		R_t–Temp	R_t–Precip	R_t–Temp	R_t–Precip	R_t–Temp	R_t–Precip
Setting 1: IC = 4 days, exIF = 30 days, W_t = 0.5	5-week lag	0.33	0.36	0.02	0.27	0.33	0.27
	4-week lag	0.25	0.08	**–0.19**	**0.29**	0.28	0.08
	3-week lag	**0.54**	0.09	–0.13	–0.22	**0.54**	0.05
	2-week lag	0.46	0.30	–0.06	–0.01	0.46	0.35
	1-week lag	0.39	0.12	–0.13	0.02	0.40	0.08
	On the week	0.32	**0.58**	–0.16	–0.18	0.33	**0.59**
		T7		T18		Guangzhou	
		R_t–Temp	R_t–Precip	R_t–Temp	R_t–Precip	R_t–Temp	R_t–Precip
Setting 2: IC = 10 days, exIF = 20 days, W_t = 0.25	5-week lag	0.28	–0.02	0.26	0.03	0.41	0.26
	4-week lag	**0.34**	0.20	**0.29**	0.22	0.51	0.36
	3-week lag	0.17	0.19	0.22	0.16	**0.63**	0.17
	2-week lag	0.22	**0.21**	0.19	**0.68**	0.62	**0.51**
	1-week lag	0.33	0.04	0.22	0.24	0.61	0.20
	On the week	0.33	0.00	0.26	–0.07	0.62	0.31

Note: IC = incubation period; exIF = extended infectious period; Temp = weekly mean temperature; Precip = weekly total precipitation. The values in the table are the Pearson correlation coefficients. The results show the lag effect of climate on the disease; for example, 1-week lag means that the correlation was calculated using the climate time series one week ahead of the R_t time series. The results of the two largest trees from each sample parameter setting are presented here.

examples to demonstrate a spatialized detection of the dengue–environment association. For the detection, we ran the Pearson correlation analysis between the twenty-four-week time series of climate factors and R_t. To detect the lag effect of the climate on the disease, we temporally shifted the correspondence between the two time series by different numbers of weeks.

At the global scale, we calculated the overall weekly average temperature and total precipitation for the entire city of Guangzhou by averaging the climate values of those pixels that fall into Guangzhou's boundary. For the climate values of an individual tree, we first created a 5-km buffer of all cases on the tree to define the geographic territory of the tree and then averaged the climate values of those pixels that fall into the tree territory. For the pixel-wise R_t, we directly link the time series of R_t of each pixel to the time series of a climate factor of the same pixel. The pixel-wise analysis generates a continuous surface of the correlation coefficient. Considering the statistical power, only those cells that have R_t values for more than twelve weeks (not necessarily consecutive) are included for the correlation analysis.

Table 5 shows the correlation analysis results for the global and tree-wise scales. Those fairly high correlation coefficient values suggest that a three- to four-week lag for temperature and a zero- to two-week lag for precipitation have considerable influence on the dengue epidemic. The difference between the tree-specific situations and their corresponding overall situations is distinctive, revealing that tree-specific association with the climate cannot be fully represented by the global correlation coefficient.

Figure 12 shows the raster maps of the pixel-wise R_t–climate correlation coefficient, providing detailed information about R_t–climate correlation. The classification facilitates identification of positive- and negative-correlated areas, their spatial distributions, sizes, and dynamics. The maps suggest that temperature is generally positively related with R_t, and the strongest association occurs at the no-lag and one-week-lag correspondences, whereas the effect of precipitation on R_t is more complicated but a two-week lag can be discerned. These findings are not completely consistent with the results of the tree-wise and the global scales, indicating that pixel-wise analysis can produce new information.

Discussion and Conclusion

An epidemic forest represents an epidemic within a particular spatiotemporal domain that starts from one or more primary cases. It explicitly models and represents the transmission linkage between a parent case and a child case. Constructing an epidemic

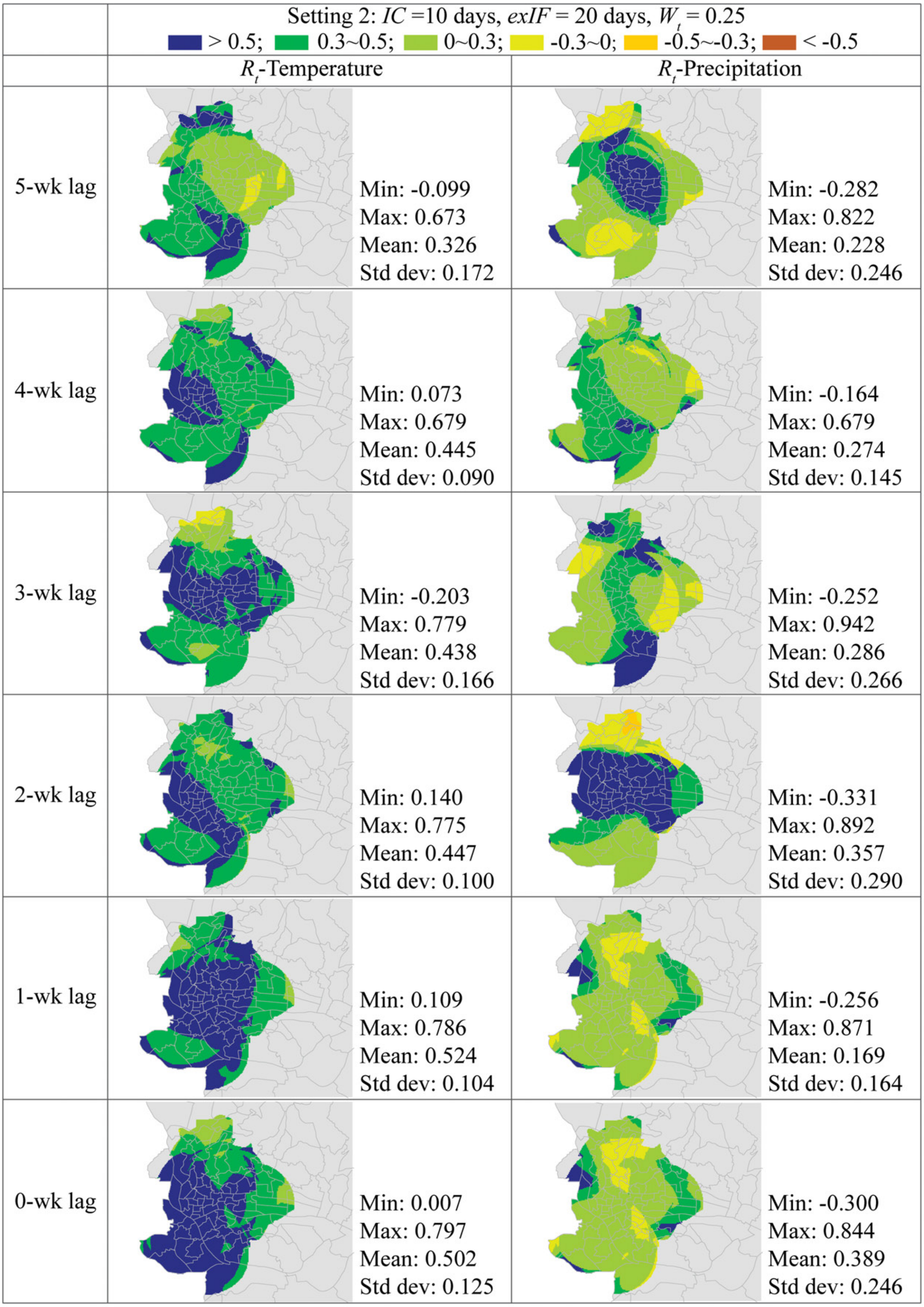

Figure 12. Pixel-wise correlation between R_t and climate factors for the 2013 dengue epidemic in Guangzhou, using Setting 2 as an example. The pixel value is the Pearson correlation coefficient between the R_ts of twenty-four epidemic weeks at that pixel and their corresponding climate values. The correlation was evaluated between R_ts and climate values of different lags. For example,1-wk lag means that the climate values are one week ahead of the R_ts values.

forest should be an effective approach to taking advantage of detailed spatial and temporal information of cases of communicable diseases when such data are available. Most important, it is a straightforward way to spatialize R_t, a widely used measurement in modeling the spread of communicable diseases in epidemiology. This article provides a formalization to the epidemic forest approach, particularly for vector-borne diseases. The formalization intends to systemize the components, parameters, procedures, analysis, and information utilization of this approach.

Another major point this article is trying to make is that localizing R_t, especially estimating pixel-wise R_t, is the basic disease mapping process for communicable diseases, corresponding to the basic process of mapping incidence or prevalence of chronic diseases. Whereas mapping a chronic disease is essentially spatializing the incidence (or prevalence; both can be simplistically interpreted as the cases-to-population ratio), mapping a communicable disease should be spatializing R_t (can be interpreted as the child-to-parent ratios). Once this understanding is established, those concepts and techniques in mapping chronic diseases, such as kernel methods (Shi 2010), can be adapted to mapping communicable diseases.

In the case study presented in this article, we constructed epidemic trees for the 2013 dengue fever epidemic in Guangzhou City, China. This study is among the earliest that build epidemic forests for human diseases. Owing to the endemic nature of dengue disease in mainland China, we were able to readily identify the primary cases and build multiple trees (i.e., the forest) for one epidemic. Through this study, we have several findings that can be summarized as follows. First, there is almost always a dominant tree, taking 42 percent to 88 percent of all cases, under different parameter settings, which suggests that a dengue epidemic is likely to have a superspreader. Identifying the superspreader and its epidemic tree in the early stage of an epidemic can be crucial for the prevention and control of the epidemic. The epidemic forest method apparently has a great advantage over traditional nonspatial epidemic modeling in being able to identify the superspreader. Second, each epidemic tree has its own structural, spatial and temporal, and epidemiological properties that cannot be fully represented by the overall information about the entire study area. Third, by calculating and analyzing R_t at three spatial scales, including global, tree-wise, and pixel-wise, we found that the three types of R_ts convey different information: The global R_t is for characterizing the overall situation of the entire study area; the tree-wise R_t is essentially process oriented, characterizing a particular spreading process; and the pixel-wise R_t is essentially location oriented, depicting the situation of each and every small area. In turn, the correlations between the R_ts at different scales and climates also convey global, process oriented, and location oriented information.

The following paragraphs summarize the limitations of this study and our future research agenda.

1. Our process for constructing the epidemic forest is simplistic and deterministic; due to the uncertainty in establishing the parent–child linkage, stochastic processes are necessary (Keeling et al. 2001; Sainudiin and Welch 2016).
2. We are aware that the use of patients' residential locations in evaluating the spatial relationship between the patients might not be entirely suitable and might to some extent distort the constructed epidemic trees. In this study, though, we were limited by data availability and not able to further improve the determination of parent–child linkage by incorporating additional information. Although we considered that the residential locations can serve the purpose of proposing and illustrating the conceptual framework and implementation of the methodology, a more reasonable and realistic representation of the patient's location is critical to this methodology. We will expand current implementation by incorporating information about the social network (Kwan 2007; Bian et al. 2012; Emch et al. 2012; Wen, Lin, and Fang 2012), transportation (Colizza et al. 2006; Nakata and Röst 2015), individual and population mobility (Kwan 1999; Mao and Bian 2010; Mao et al. 2016), and public health intervention (J.-F. Wang et al. 2006) to improve the evaluation of connection between individuals.
3. Although the spatiotemporal distance might dominate the transmission of infectious diseases, person-to-person and human–vector contacts feature uncertainty and an occasional contact might cause a quite different route of transmission, exemplified by the transmission caused by the first SARS case in late 2002 (Meng et al. 2005; J.-F. Wang et al. 2006). The algorithm implemented in this study, which is solely based on simple spatiotemporal information about patients, including the residential location and onset time, might be overly simplistic for modeling the complicated transmission process. Nevertheless, first, the simplicity of the algorithm and data provide

researchers and practitioners an easy way to gain basic knowledge of spatiotemporal characteristics of the epidemic when other information is not available; second, the methodology provides an open framework that is capable of incorporating and actually taking advantage of auxiliary data and methods. In particular, data on transportation, human mobility, and genetic analysis can be used to improve the quality of the epidemic forest and verify its accuracy. Utilization of big data, genetic data, and machine learning techniques in construction of epidemic forests is on our future research agenda.

4. In this study, we applied constant parameter values to the entire study area. Although this global setting is simple and easy, it could be confounding in either spatial or temporal or both dimensions. The accuracy of the epidemic forest might be improved by localizing the parameter setting through incorporating a priori knowledge and observed data. To this end, the analysis of spatial stratified heterogeneity is a promising strategy to balance the spatial precision and statistical power (J.-F. Wang, Zhang, and Fu 2016).
5. The analysis of disease-environment associations implemented in the case study is also simplistic. The influence of environmental factors on communicable diseases can be particularly complicated and nonlinear, requiring more sophisticated methods to be employed (J.-F. Wang et al. 2010; Yin and Wang 2017; Liao et al. 2018; Kwan 2018).
6. Meanwhile, with observed (true) data, the parameters in constructing the epidemic forest can be calibrated through machine learning or other techniques. The most reliable determination of transmission linkage is through genetic analysis. We have been working on collecting this kind of genetic analysis data.

Through this study, we explored and intend to demonstrate what can be done to spatialize epidemiological modeling, what information can be extracted from the output of this spatialization, and then how to use the extracted information.

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